

CS1231S: Discrete Structures
Counting and Probability II and Graphs I
 (Week 12: 8 – 12 April 2024)
Answers

1. Find the term independent of x in the expansion of $\left(2x^2 + \frac{1}{x}\right)^9$.

Answer:

Recall the Binomial Theorem:

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n-1} a^1 b^{n-1} + b^n$$

We have $a = 2x^2$, $b = \frac{1}{x}$, $n = 9$.

The general term in the expansion of $(a + b)^n$ is given by:

$$\binom{n}{r} a^{n-r} b^r = \binom{9}{r} (2x^2)^{9-r} \left(\frac{1}{x}\right)^r = \binom{9}{r} 2^{9-r} \cdot x^{18-2r} \cdot x^{-r} = \binom{9}{r} 2^{9-r} \cdot x^{18-3r}$$

For this term to be independent of x , we must have $18 - 3r = 0$, or $r = 6$.

Therefore, the term independent of x is

$$\binom{9}{6} 2^{9-6} = 84 \times 2^3 = \mathbf{672}$$

2. Let's revisit Tutorial #9 Question 4:

Given n boxes numbered 1 to n , each box is to be filled with either a white ball or a blue ball such that at least one box contains a white ball and boxes containing white balls must be consecutively numbered. What is the total number of ways this can be done?

Last week, the answer given was

For k ($1 \leq k \leq n$) consecutively numbered boxes that contain white balls, there are $n - k + 1$ ways. Therefore, total number of ways is $\sum_{k=1}^n (n - k + 1) = \sum_{k=1}^n k = \frac{n(n+1)}{2}$.

Now, let's use another approach to solve this problem. Draw crosses on the side of the boxes as shown below. How do you use these crosses?



Answer:

The task is similar to choosing two out of the $n + 1$ crosses to mark the start and end of the consecutively numbered boxes that contain white balls.

This is $\binom{n+1}{2}$, which is also equal to $n(n + 1)/2$.

3. [AY2020/21 Semester 2 Exam Question]

On a die there are 6 numbers. We call 4, 5, 6 the big numbers and 1, 2, 3 the small numbers. Given a loaded die in which the probability of rolling any fixed big number is twice the probability of rolling any fixed small number, answer the following questions.

- What is the probability of rolling a 6? Write your answer as a single fraction.
- If two such loaded dice are rolled, what is the expected value of the maximum of the two dice? Write your answer as a single fraction.

Answer:

Probability of $\frac{1}{9}$ to roll a 1, 2 or 3; probability of $\frac{2}{9}$ to roll a 4, 5 or 6.

Maximum of 1: 1 way $\rightarrow (1,1)$ with probability $\frac{1}{81}$.

Maximum of 2: 3 ways $\rightarrow (1,2), (2,1), (2,2)$ with probability $\frac{3}{81}$.

Maximum of 3: 5 ways $\rightarrow (1,3) \times 2, (2,3) \times 2, (3,3)$ with probability $\frac{5}{81}$.

Maximum of 4: 7 ways $\rightarrow (1,4) \times 2, (2,4) \times 2, (3,4) \times 2, (4,4)$ with probability $\frac{12}{81} + \frac{4}{81} = \frac{16}{81}$.

Maximum of 5: 9 ways $\rightarrow (1,5) \times 2, (2,5) \times 2, (3,5) \times 2, (4,5) \times 2, (5,5)$ with probability $\frac{12}{81} + \frac{12}{81} = \frac{24}{81}$.

Maximum of 6: 11 ways $\rightarrow (1,6) \times 2, (2,6) \times 2, (3,6) \times 2, (4,6) \times 2, (5,6) \times 2, (6,6)$ with probability $\frac{12}{81} + \frac{20}{81} = \frac{32}{81}$.

Expected value = $\left(\frac{1}{81} \times 1\right) + \left(\frac{3}{81} \times 2\right) + \left(\frac{5}{81} \times 3\right) + \left(\frac{16}{81} \times 4\right) + \left(\frac{24}{81} \times 5\right) + \left(\frac{32}{81} \times 6\right) = \frac{398}{81}$.

4. An urn contains five balls numbered 1, 2, 2, 8 and 8. If a person selects a set of two balls at random, what is the expected value of the sum of the numbers on the balls?

Answer:

Let $2_a, 2_b$ denote the two balls with the number 2, and $8_a, 8_b$ the two balls with the number 8.

The list below shows the sums of the numbers on the balls in each set:

- Sum of 3: $\{1, 2_a\}, \{1, 2_b\}$.
- Sum of 4: $\{2_a, 2_b\}$.
- Sum of 9: $\{1, 8_a\}, \{1, 8_b\}$.
- Sum of 10: $\{2_a, 8_a\}, \{2_a, 8_b\}, \{2_b, 8_a\}, \{2_b, 8_b\}$.
- Sum of 16: $\{8_a, 8_b\}$.

Expected value = $3 \times \left(\frac{2}{10}\right) + 4 \times \left(\frac{1}{10}\right) + 9 \times \left(\frac{2}{10}\right) + 10 \times \left(\frac{4}{10}\right) + 16 \times \left(\frac{1}{10}\right) = 8.4$.

Or, we could compute the expected value of the number when one ball is picked:

$1 \times \left(\frac{1}{5}\right) + 2 \times \left(\frac{2}{5}\right) + 8 \times \left(\frac{2}{5}\right) = 4.2$.

Then apply **linearity of expected value** to get $2 \times 4.2 = 8.4$.

5. [AY2020/21 Semester 1 Exam Question]

In 1995, Mars[®] introduced the blue M&M's to replace the tan M&M's. Before that, the colour mix in a bag of plain M&M's was 30% brown, 20% yellow, 20% red, 10% green, 10% orange and 10% tan. After the replacement, the colour mix was 24% blue, 16% green, 20% orange, 14% yellow, 13% red and 13% brown.



Your friend Aiken, being a big fan of M&M's, has two bags of plain M&M's, one from 1994 and the other from 1996. He won't tell you which is which, but he gives you one M&M from each bag. One is yellow and one is green. What is the probability that the **yellow M&M came from the 1994 bag**, assuming the M&M's are picked randomly.

You may express your answer as a fraction or a decimal number correct to 3 significant digits

Answer:

Let A : Bag 1 from 1994 and bag 2 from 1996, and B : Bag 1 from 1996 and bag 2 from 1994.

$$P(A) = P(B) = \frac{1}{2}.$$

Let E : yellow M&M from bag 1, green M&M from bag 2.

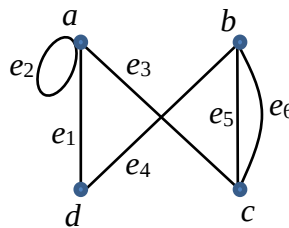
Then $P(E|A) = 0.2 \times 0.16 = 0.032$ (yellow from 1994 and green from 1996)

and $P(E|B) = 0.14 \times 0.1 = 0.014$ (yellow from 1996 and green from 1994)

By Bayes' Theorem,

$$P(A|E) = \frac{P(E|A) \cdot P(A)}{P(E|A) \cdot P(A) + P(E|B) \cdot P(B)} = \frac{0.032 \times \frac{1}{2}}{0.032 \times \frac{1}{2} + 0.014 \times \frac{1}{2}} = \frac{0.016}{0.023} = \frac{16}{23} \text{ or } 0.696.$$

6. Given the graph shown below:



(a) Give the adjacency matrix A for the graph, with vertices in the order a, b, c, d .

(b) Compute A^0 , A^2 and A^3 .

(c) How many walks of length 2 are there from a to b ? From c to itself? List out all the walks.

(d) How many walks of length 3 are there from a to c ? List out all the walks.

Answers:

(a)

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$(b) \quad A^0 = I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad A^2 = \begin{bmatrix} 3 & \mathbf{3} & 1 & 1 \\ 3 & 5 & 0 & 0 \\ 1 & 0 & \mathbf{5} & 3 \\ 1 & 0 & 3 & 2 \end{bmatrix}; \quad A^3 = \begin{bmatrix} 5 & 3 & \mathbf{9} & 6 \\ 3 & 0 & 13 & 8 \\ 9 & 13 & 1 & 1 \\ 6 & 8 & 1 & 1 \end{bmatrix}$$

(c) There are **3** walks of length 2 from a to b :

$$\langle a \rightarrow c \xrightarrow{e_5} b \rangle, \langle a \rightarrow c \xrightarrow{e_6} b \rangle, \langle a \rightarrow d \rightarrow b \rangle.$$

There are **5** walks of length 2 from c to itself:

$$\langle c \rightarrow a \rightarrow c \rangle, \langle c \xrightarrow{e_5} b \xrightarrow{e_5} c \rangle, \langle c \xrightarrow{e_5} b \xrightarrow{e_6} c \rangle, \langle c \xrightarrow{e_6} b \xrightarrow{e_5} c \rangle, \langle c \xrightarrow{e_6} b \xrightarrow{e_6} c \rangle.$$

(d) There are **9** walks of length 3 from a to c :

1 way via a first: $\langle a \rightarrow a \rightarrow a \rightarrow c \rangle$.

0 way via b first.

5 ways via c first: $\langle a \rightarrow c \rightarrow a \rightarrow c \rangle, \langle a \rightarrow c \xrightarrow{e_5} b \xrightarrow{e_5} c \rangle, \langle a \rightarrow c \xrightarrow{e_5} b \xrightarrow{e_6} c \rangle, \langle a \rightarrow c \xrightarrow{e_6} b \xrightarrow{e_5} c \rangle, \langle a \rightarrow c \xrightarrow{e_6} b \xrightarrow{e_6} c \rangle$.

3 ways via d first: $\langle a \rightarrow d \rightarrow a \rightarrow c \rangle, \langle a \rightarrow d \rightarrow b \xrightarrow{e_5} c \rangle, \langle a \rightarrow d \rightarrow b \xrightarrow{e_6} c \rangle$.

7. A lady hosted a party of n ($n \geq 2$) people (including herself). At the party, various friends met and some of them shook hands with each other. The thoughtful host made sure that she shook hands with everyone at the party.

Prove that there are at least two people who have shaken hands the same number of times.

Answer:

Model this with a graph $G = (V, E)$ where each person is a vertex in V and there is an undirected edge $\{x, y\} \in E$ whenever x and y shook hand at the party.

1. Let h be the host vertex. Then $\deg(h) = n - 1$. (given by the problem)
2. For other vertices v , $\deg(v) > 0$. (shook hand with host h)
3. So all n vertices have $\deg(v)$ in the range $[1..(n - 1)]$.
4. Hence, at least two vertices have the same degree. (by the Pigeonhole Principle)

Hence at least two people have the *same* number of handshakes.

8. [AY2015/16 Semester 1 Exam Question]

Let $A = \{1, 2, 3, 4\}$. Since each element of $P(A \times A)$ is a subset of $A \times A$, it is a binary relation on A . ($P(S)$ denotes the powerset of S .)

Assuming each relation in $P(A \times A)$ is equally likely to be chosen, what is the probability that a randomly chosen relation is (a) reflexive? (b) symmetric?

Can you generalize your answer to any set A with n elements?

Answers:

- (a) $1/16$ (b) $1/64$

For the general case, let $|A| = n$. In general, we need to consider all possible n^2 pairs. Thus, the total number of possible combinations, i.e. the number of elements in $P(A \times A)$, is 2^{n^2} .

- (a) To count the number of reflexive relations, note that all the pairs (a, a) for all $a \in A$ must be in the relation, so these n pairs are fixed. We are then free to choose to include or not any of the $(n^2 - n)$ remaining pairs. This gives us:

$$\frac{2^{n^2-n}}{2^{n^2}} = \frac{1}{2^n}$$

- (b) For symmetric relations, if some pair (a, b) is in the relation, then (b, a) must also be in the relation. Hence, either (a, b) and (b, a) are both included in the relation, or both are not. This gives us $\frac{n^2+n}{2}$ pairs to choose to include or not, giving us:

$$\frac{2^{\frac{n^2+n}{2}}}{2^{n^2}} = \frac{1}{2^{\frac{n^2-n}{2}}}$$

(Refer to the slides uploaded for more details.)

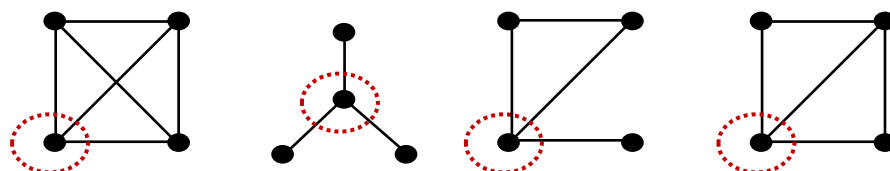
9. [AY2021/22 Semester 1 Exam]

A set of vertices, D , in an undirected simple graph is said to be a **dominating set** if every vertex not in D is adjacent to at least one vertex in D . A **minimal dominating set** is a dominating set such that none of its proper subsets are dominating sets.

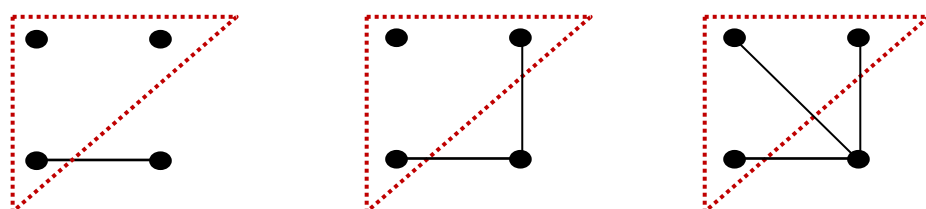
- (a) Draw two non-isomorphic simple graphs with four vertices that have minimal dominating sets of size one. Highlight the vertices in the minimal dominating set in your graphs.
- (b) Draw two non-isomorphic simple graphs with four vertices that have minimal dominating sets of size three. Highlight the vertices in the minimal dominating set in your graphs.

Answers:

- (a) We accept any two of the following answers. Dominating set shown in dotted circle.



- (b) We accept any two of the following answers. Dominating set shown in dotted triangle.



10. Let G be any simple graph with 6 vertices. Prove that G or its complementary graph $\bar{G}$ contains a triangle. (This is similar to the problem of showing that in any group of 6 people, there must be either 3 mutual friends or 3 mutual strangers. But for this question, please use the graph formulation.)

Answer:

1. Let G be a simple graph with 6 vertices. (eg: Fig. 10a)
2. In K_6 , the complete graph on 6 vertices, we colour the edges in G **black**, and those not in G **red**. Namely, the **red** edges are those in the complement graph $\bar{G}$. (See Figs 10a-c.)

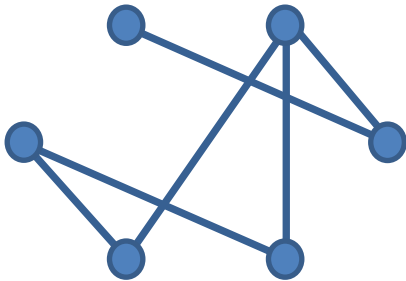


Fig. 10a. The graph

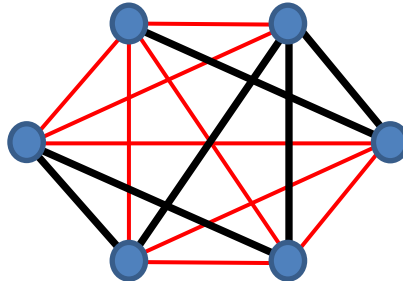


Fig. 10b. The coloured K_6 graph.

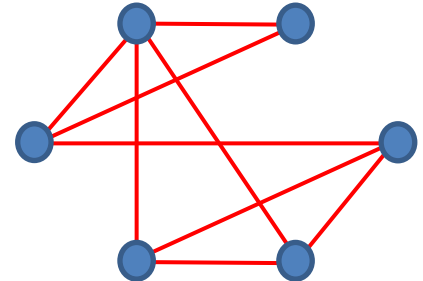


Fig. 10c. The complementary graph

3. In this coloured K_6 , the edges in G are coloured black while those in $\bar{G}$ are red. We want to prove that this coloured K_6 contains a black triangle (cycle of length 3) or a red triangle.
4. Let v be an arbitrary vertex of the coloured K_6 .
 - 4.1. There are 5 edges incident to v (since K_6 is a complete graph), coloured black or red.
 - 4.2. Therefore, (at least) 3 of them have the same colour C , by the Generalized PHP. (In Figure 10d, 3 edges incident to v are red.)
 - 4.3. WLOG, let the colour C be red and the red edges be $\{v, u_1\}$, $\{v, u_2\}$, $\{v, u_3\}$. (Fig 10d).

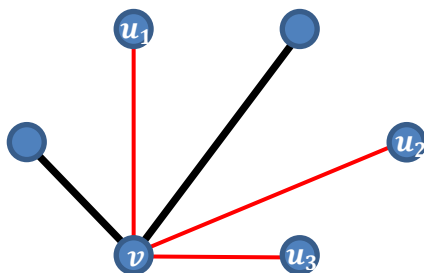


Fig. 10d

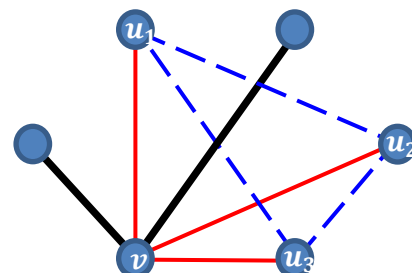


Fig. 10e

- 4.4. **Case 1:** If at least one of the edges $\{u_1, u_2\}$, $\{u_2, u_3\}$ or $\{u_1, u_3\}$ is red. If $\{u_x, u_y\}$ is the edge that is red, then we have a red triangle $\{v, u_x, u_y\}$. (See Fig. 10e.) This is a triangle in $\bar{G}$.
- 4.5. **Case 2:** If all the three edges $\{u_1, u_2\}$, $\{u_2, u_3\}$ or $\{u_1, u_3\}$ are black. Then, we have a black triangle $\{u_1, u_2, u_3\}$. (See Fig. 10e.) This is a triangle in G .
5. Therefore, G or $\bar{G}$ contains a triangle.

[Note: For practice, students may want to also write out the case for “WLOG, let C be black”. And also modify Figures 10d and 10e, accordingly.]

The following definitions are used in Question 11.

A relation $<$ on a set A is said to be **irreflexive**, if and only if, $\forall a \in A, (a \not< a)$.

Alternate definition of anti-symmetry: $\forall x, y, (x \neq y) \Rightarrow ((x, y) \in R) \Rightarrow ((y, x) \notin R)$.

A relation is a **strict partial order** if and only if it is irreflexive, antisymmetric and transitive.

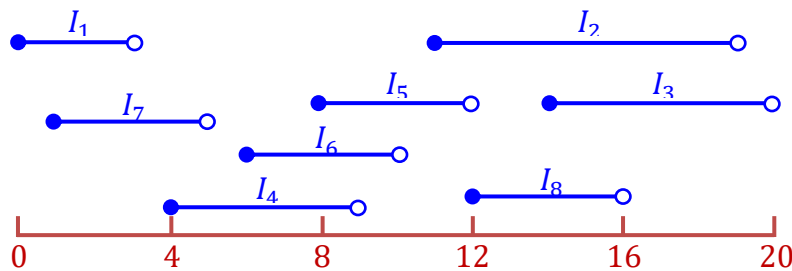
Let $\leq$ be a strict partial order on a set A . A subset C of A is called **chain** if and only if each pair of *distinct* elements in C is comparable, that is, $\forall a, b \in C (a \neq b) \Rightarrow (a \leq b \vee b \leq a)$.

A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain.

11. You are given a set of n jobs $J = \{J_1, J_2, J_3, \dots, J_n\}$. Each job J_k is represented by the interval $I_k = [s_k, e_k)$ where s_k is the *start time* and e_k (where $s_k < e_k$) is the *end time* of the job, for $k = 1, 2, \dots, n$. An instance of this problem with $n = 8$ is shown below.

Instance: $n = 8$, and $\{J_1, J_2, J_3, \dots, J_8\}$, and

$$\begin{array}{llll} I_1 = [0, 3), & I_2 = [11, 19), & I_3 = [14, 20), & I_4 = [4, 9) \\ I_5 = [8, 12), & I_6 = [6, 10), & I_7 = [1, 5), & I_8 = [12, 16), \end{array}$$



Define a relation $<$ on J by: $(J_x < J_y) \Leftrightarrow (e_x \leq s_y)$

Namely, $J_x < J_y$ means that the job J_x is *to the left of* the job J_y .

- (a) Show that the relation $<$ is a strict partial order.

Answer

To show $<$ is irreflexive:

1. Consider any job $J_k \in J$ with interval $I_k = [s_k, e_k)$.
2. Then $e_k > s_k$ (by definition of the jobs J_k)
3. Hence $(J_k \not< J_k)$. (by definition of $<$)
4. Hence, $<$ is irreflexive. (by definition of irreflexivity)

To show $<$ is anti-symmetric: (Using the alternative definition of anti-symmetry.)

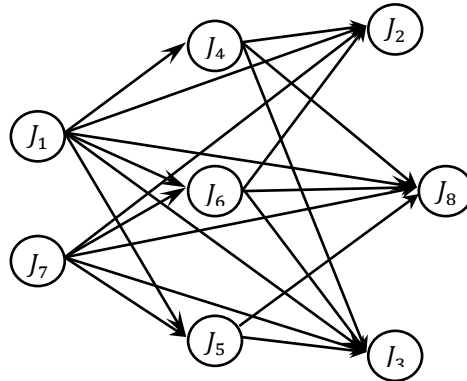
1. For any 2 *distinct* jobs J_x, J_y with interval $I_x = [s_x, e_x)$ and $I_y = [s_y, e_y)$, respectively,
2. Suppose $(J_x < J_y)$. Then $(e_x \leq s_y)$. (by definition of $<$)
3. We have $(e_y > s_y)$ and $(e_x > s_x)$. (by definition of the jobs J_x, J_y)
4. Then, $e_y > s_y \geq e_x > s_x \Rightarrow e_y > s_x \Rightarrow e_y \not\leq s_x$. (by 2 and 3)
5. Hence $(J_y \not< J_x)$. Hence, $<$ is anti-symmetric.

To show $<$ is transitive:

1. For any jobs $J_x, J_y, J_z \in J$, suppose $(J_x < J_y)$ and $(J_y < J_z)$.
2. Then $(e_x \leq s_y)$ and $(e_y \leq s_z)$. (by definition of $<$)
3. We have $(s_y < e_y)$. (by definition of the job J_y)
4. Then, $e_x \leq s_y < e_y \leq s_z \Rightarrow e_x \leq s_z$. (by 2 and 3)
5. Hence $(J_x < J_z)$. Hence, $<$ is transitive.

(b) Draw the graph $G = (J, <)$ of the relation $<$ for this instance given above.

Answer:



You want to assign all the jobs to workers so as to *minimize* the number of workers deployed. Each worker can only work on one job at a time. Each worker can do any number of jobs as long as the jobs do not overlap in time. (You can assume that a worker can (if necessary) start on a new job immediately after finishing a previous job. In the instance above, jobs T_5 and T_8 can be assigned to the same worker, if necessary.) This problem can be solved in parts (c) and (d) as follows:

(c) Give a maximal chain C in the graph G . Argue that the jobs in C can be assigned to a single worker.

Answer:

Many possible answers. For example, $C = \{J_1, J_4, J_2\}$, $C = \{J_1, J_4, J_3\}$, etc.

Every chain consists of jobs that do not overlap in time, and can be assigned to a single worker.

(d) Now, partition the vertices in G into a *minimum* number of vertex-disjoint chains. For each chain, we assign a worker to do all the jobs in that chain. What is the minimum number of workers needed? Show the jobs assigned to each worker (W1, W2, etc.).

Answer:

Partition: $J = \{\{J_1, J_4, J_2\}, \{J_7, J_6, J_8\}, \{J_5, J_3\}\}$.

Minimum number of workers needed is 3. Job assignment is given below:

W1: $\{J_1, J_4, J_2\}$; W2: $\{J_7, J_6, J_8\}$; W3: $\{J_5, J_3\}$.

(Note: There are many possible answers with 3 workers.)